

Industry, Household & Children: A Multi-Perspective Analysis of U.S. Poverty

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1. Overview/Abstract

This study looks at how the types of jobs in different U.S. counties affect poverty, using data from the American Community Survey (ACS) for 2015 and 2017. The results show that poverty is not spread evenly across the country. Understanding what causes these differences is important for policymakers who want to design better economic programs. The main question is whether the mix of jobs in a county is connected to differences in poverty rates and income.

The ACS is a yearly survey from the U.S. Census Bureau that gives detailed information about people and jobs in different areas. For this study, the data includes over 10,000 counties and covers poverty rates, median income, employment, and job types. The analysis found that counties with more professional jobs usually have lower poverty and higher incomes. Counties with more production and construction jobs tend to have higher poverty. There are clear regional differences, with some areas having more high-poverty counties. From 2015 to 2017, counties that shifted toward more skilled jobs saw some economic improvement.

These results suggest that policymakers should help people get access to higher-skilled jobs. Investing in education and training can help reduce differences between regions. The interactive dashboard created for this project lets users explore these patterns and supports decisions based on data.

2. Introduction

Economic inequality and poverty are still big problems in the United States, and they vary a lot from county to county. Even though national numbers show the economy is growing, these averages can hide local differences in income, jobs, and living conditions. To create better policies, it is important to find out what causes poverty in different places. One key factor is the types of jobs available in each area.

This study analyzes how different occupational sectors (professional, service, construction, and production) relate to poverty rates and income levels across U.S. counties. It aims to identify patterns explaining why some regions experience persistently higher poverty than others, even during economic growth.

The primary stakeholders for this analysis are as follows:

- Local and state policymakers, responsible for designing workforce development and economic measures to reduce poverty
- Urban and regional planners, who require an understanding of financial disparities when allocating resources
- Workforce development agencies, which can tailor training programs to local labor market needs
- Social and economic researchers, who study inequality and labor market dynamics

3. Problem statement

Despite overall economic growth in the United States, poverty remains unevenly distributed across counties, showing that macroeconomic improvements do not benefit all communities equally. This raises a critical question: which organizational factors drive these persistent local disparities in poverty and income?

A central hypothesis is that the occupational composition of a county's labor force significantly influences its economic outcomes. Counties with more lower-wage, lower-skill occupations, such as production,

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construction, and certain service jobs, may have higher poverty rates and lower incomes. Conversely, counties with a larger share of professional and managerial occupations are expected to show greater economic resilience and lower poverty rates.

However, this relationship is not fully understood, especially its variation across regions and evolution over time. It also remains unclear whether changes in occupational structure are linked to measurable improvements or declines in poverty over several years.

This study aims to analyze and quantify the relationship between occupational sectors and poverty across U.S. counties using data from 2015 and 2017. Specifically, it examines patterns linking job composition to economic outcomes, identifies geographic disparities, and evaluates whether shifts in occupational structure correspond to changes in poverty levels.

4. Data used

This project uses four datasets from the U.S. Census Demographic Data, covering both county-level and census tract-level data for 2015 and 2017.

- `acs2015_census_tract_data.csv`: Census tract-level data for the U.S., including DC and Puerto Rico (2015).
- `acs2015_county_data.csv`: County-level data for the U.S., including DC and Puerto Rico (2015).
- `acs2017_census_tract_data.csv`: Census tract-level data for the U.S., including DC and Puerto Rico (2017).
- `acs2017_county_data.csv`: County-level data for the U.S., including DC and Puerto Rico (2017).

The combined county-level panel dataset contains over 6,400 rows and includes the following key attributes.

Occupational Categories:

- Professional — Share of workforce in management, business, science, and arts
- Service — Share of workforce in service occupations
- Office — Share of workforce in sales and office occupations
- Construction — Share of workforce in natural resources, construction, and maintenance
- Production — Share of workforce in production, transportation, and material moving

Economic Indicators:

- Poverty — County-level poverty rate (%)
- ChildPoverty — Child poverty rate (%)
- Income — Median household income (USD)
- IncomePerCap — Per capita income (USD)
- Unemployment — Unemployment rate (%)

Geographic Identifiers:

- CountyId / FIPS — Unique county identifier in 5-digit format for GeoJSON mapping
- State, County — State and county name

The following preprocessing steps were applied consistently to both county and tract datasets.

Column names were unified across datasets to ensure consistent joins and comparisons. Specifically, `CensusId` was renamed to `CountyId` and `Citizen` was renamed to `VotingAgeCitizen` in the 2015 county

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dataset, and CensusTract was renamed to TractId in the 2015 tract dataset. Additionally, the string "County" was stripped from county name entries in the 2017 datasets to normalize naming conventions.

The CountyId field was converted from integer to a zero-padded 5-digit string to match the GeoJSON geographic boundary file required for choropleth mapping. Without this step, county-level map visualizations would fail to join correctly on FIPS codes.

A Year column was added to each dataset before concatenation. The two annual datasets were merged using `pd.concat()` to form a single longitudinal panel, enabling year-over-year comparisons across all visualizations.

For all key analytical columns, missing values were filled with the column-level mean, preserving sample size while minimizing distributional distortion. This approach was chosen to avoid reducing the dataset size, which would have disproportionately affected smaller, data-sparse rural counties.

5. Exploratory Analysis

Link to our analysis github: [link](#)

The analytical process followed a structured, top-down framework moving from national overview to county-level and individual-level detail.

Our analysis begins by establishing a national baseline. We first examined the overall distribution of poverty rates across all counties for both 2015 and 2017, identifying key descriptive statistics such as the mean, median, and the proportion of high-poverty counties. Alongside this, we calculated the average share of each occupational category — Professional, Service, Office, Construction, and Production — across all counties for both years, building a foundational understanding of how the American workforce is structured and how that structure relates to economic outcomes at the aggregate level.

With the national picture established, we then turned to geography. County-level and state-level poverty rates were visualized on choropleth maps, layered with the dominant occupational type in each area, to reveal whether and how poverty clusters spatially across the United States. This geographic lens allowed us to move beyond aggregate averages and identify which regions face structurally entrenched disadvantages — and whether that disadvantage aligns with particular types of local labor markets.

Having identified spatial patterns, we next investigated the statistical relationship between occupational structure and poverty more rigorously. Scatter plot analysis was conducted for each occupational category against the poverty rate, with Pearson correlation coefficients computed for both years to quantify the direction and strength of each relationship. Building on this, we constructed a full correlation heatmap between all five occupational categories and five economic outcome variables — poverty, child poverty, income, per capita income, and unemployment — providing a comprehensive and comparative view of which occupational sectors most consistently predict economic distress or prosperity.

To ground these statistical findings in a concrete policy context, we then compared occupational composition and poverty rates across the 15 highest-poverty and 15 lowest-poverty states. This side-by-side comparison made visible the structural differences in labor market composition between economically thriving and persistently struggling states, translating abstract correlations into actionable geographic and policy insights.

Finally, we extended our analysis beyond county-level averages to examine inequality at a finer geographic resolution. Census tract-level data and box plots were used to reveal the within-state poverty dispersion that county averages tend to obscure, exposing pockets of extreme deprivation hidden inside otherwise moderate-performing states. As a targeted complement to this, we identified the top 10 counties with the highest child poverty rates, highlighting the most geographically concentrated and socially

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vulnerable communities in the dataset — a dimension of poverty that aggregate statistics consistently underrepresent.

6. Dashboard Design

Link to our dashboard: [link](#)

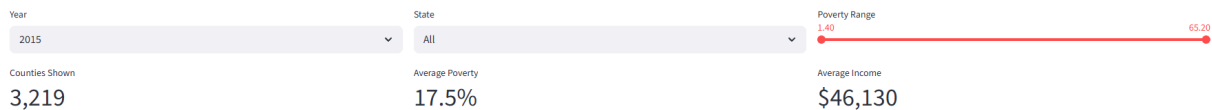
Link to our dashboard github: [link](#)

The dashboard is designed to provide an interactive and user-driven exploration of poverty patterns and their relationship with occupational structure across U.S. counties. To support flexible analysis, the interface includes several key interactive components that allow users to filter and explore the data dynamically. Besides, we use different colors of intensity to represent the severity of the situation, and for each chart we have an auto quick analysis to explain the information.

1. Name and Feature Selection

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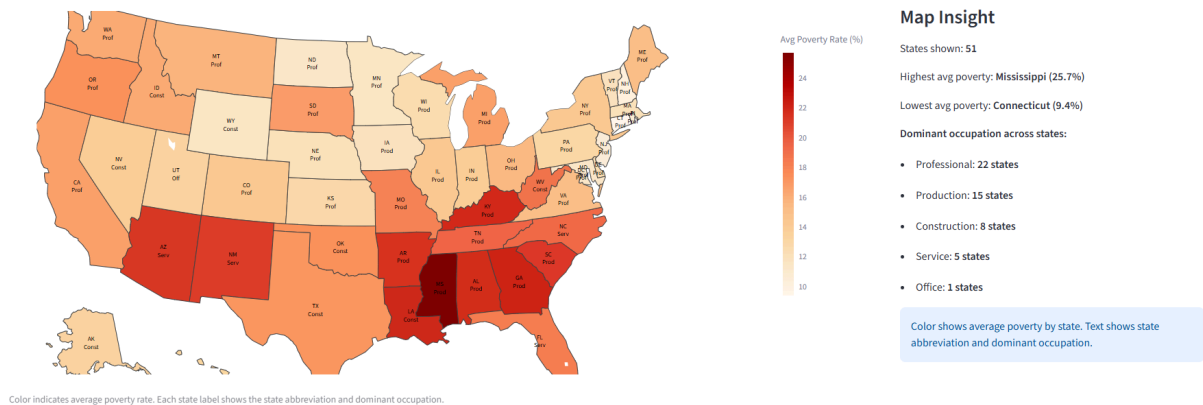
This dashboard compares county-level data from 2015 and 2017 to explore how occupational structure relates to poverty and income across U.S. counties.



In addition to enabling variable selection (e.g., year, state, and poverty range), the dashboard highlights key summary statistics using large, prominent text to improve readability and support quick insight extraction.

2. Geographic Poverty Distribution

State-Level Poverty and Dominant Occupation (2015)

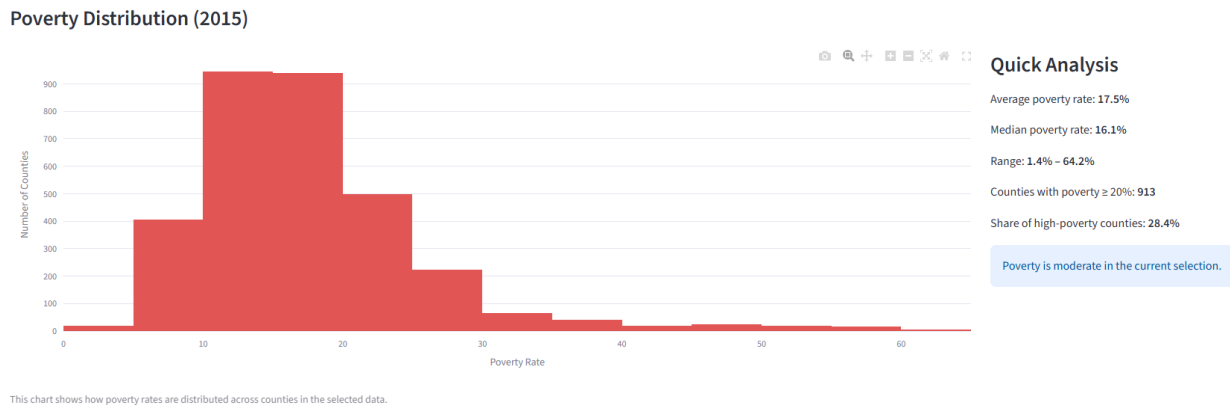


A choropleth map is used to display state-level poverty rates, with color intensity encoding the severity of poverty. This design allows users to quickly identify geographic patterns and regional disparities across

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the United States. The map shows that higher poverty rates are concentrated in the southern states, with a higher percentage of occupation of products and services.

3. Top 10 Counties by Poverty



This histogram shows the distribution of poverty rates across counties. Most counties are concentrated between 10% and 20%, indicating moderate poverty levels.

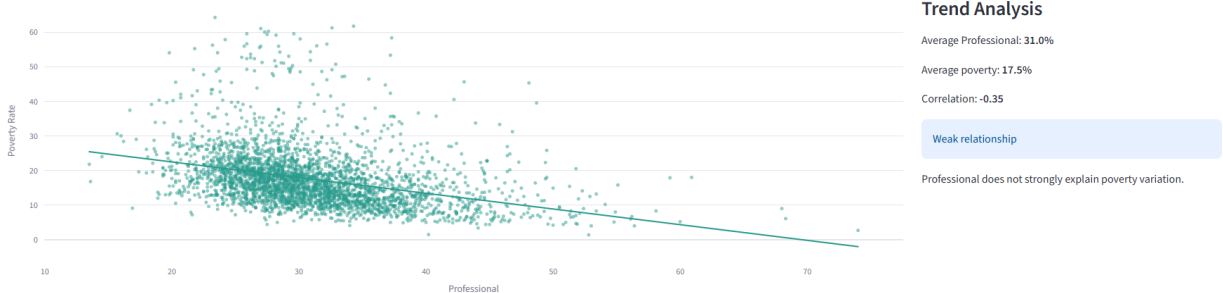
4. Occupation vs Poverty Relationship

Select Occupation for Analysis

Occupation

Professional

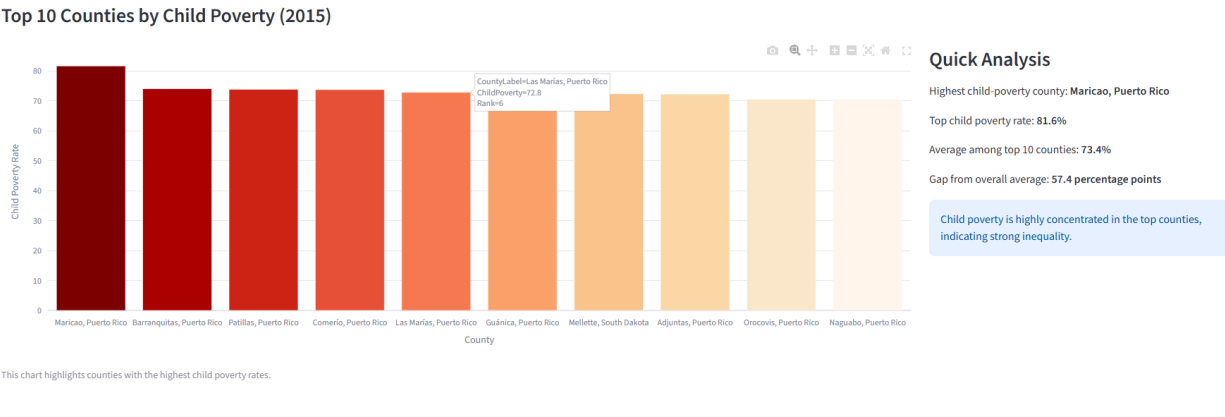
Professional vs Poverty (2015)



A scatter plot with a fitted trend line is used to examine the relationship between the share of professional occupations and poverty rates across counties. The chart shows a negative correlation, indicating that counties with a higher proportion of professional workers tend to have lower poverty rates; but for other occupations, there's actually a positive correlation.

5. Child Poverty

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A bar chart is used to highlight the top 10 counties with the highest child poverty rates, allowing for clear comparison across counties. The use of color intensity emphasizes the most severe cases, helping users quickly identify extreme values. The visualization shows that child poverty is highly concentrated in a small number of counties, with the top county reaching over 80%, significantly exceeding the overall average, indicating substantial inequality and localized economic distress.

7. Conclusion

This study analyzed the relationship between occupational structure and poverty across U.S. counties using ACS data from 2015 and 2017. The findings consistently demonstrate that occupational composition is a meaningful and stable predictor of county-level poverty outcomes.

Counties with a higher share of Professional employment exhibit lower poverty rates, higher incomes, and lower child poverty. Conversely, counties dominated by Production and Service occupations face structurally higher poverty, with this pattern persisting across both years studied. Geographic analysis confirms a spatial concentration of poverty in the Deep South and Puerto Rico, areas characterized by limited Professional sector presence and historical underinvestment.

The analysis of census tract-level data further reveals that within-state inequality is often more severe and more hidden than between-state comparisons suggest, underscoring the need for place-based policy interventions operating below the state level.